Input data for three volcanoes

Etna (Italy) · Campi Flegrei (Italy) · Kīlauea (Hawai‘i) — the ready-made inputs of the notebook *From magma compressibility to surface uplift* (EVo + dMODELS), compiled from published petrology and geodesy

How to use. In cell 4 of the notebook choose the volcano under *Start values*: the magma table (cell 5) and the reservoir table (cell 6) are filled with the values below, and any of them can be changed directly in the tables. Every value below is either taken from the cited publication or, where marked *assumption / derived*, chosen or calculated as explained in the note.

Conventions

• Volatile contents are **total budgets** (dissolved + exsolved) in wt%, supplied to EVo in saturation mode: EVo finds the saturation pressure and decompresses the closed system to the reservoir pressure. Melt-inclusion values are minima for the pre-eruptive budget (inclusions trap a melt that may already have lost CO_2).

• Major elements are volatile-free wt% with all iron as FeO_{total} (Fe_2O_3(total) × 0.8998 where the source reports Fe_2O_3). EVo normalises the composition itself. EVo solubility class: basalt 45–55, phonolite 52–63, rhyolite 65–80 wt% SiO_2 (Burgisser et al. 2015).

• β_{liquid} and ρ_{melt} affect the result only when the magma is vapour-free; with vapour present EVo supplies the melt density. The other EVo settings are those of the thesis (cell 7 of the notebook).

• Depths are those of the source centres in a flat elastic half-space. The reference surface and the pressure datum are stated for each volcano because published depths are given relative to different references.

Overview

Volcano	Magmas	Reservoirs	Main sources
Etna (Italy)	ETNA_PRIMITIVE — Primitive 2002 basalt (lapilli, 21 Nov 2002) ETNA_PONDING — Evolved 2002 magma ponding at 4–6 km b.s.l. ETNA_DEGASSED — Degassed shallow melt (2002 matrix glass)	Recharge source · McTigue (finite sphere (McTigue), 8.1 km) Discharge / ponding source · McTigue (finite sphere (McTigue), 6.1 km) Main chamber · sill (penny-shaped crack (Fialko), 6.1 km) Lava-fountain source · gas-rich (prolate spheroid (Yang), 2.35 km) Lava-fountain source · degassed (prolate spheroid (Yang), 2.35 km)	Aloisi 2018, Bonaccorso 2013, Bonaccorso 2021, Gennaro 2020, Scudero 2019, Spera 2000, Spilliaert 2006
Campi Flegrei (Italy)	CF_TRACHYTE — Trachyte, Agnano–Monte Spina type (4.1 ka) CF_TRACHYBASALT — Trachybasalt (shoshonitic parental melt, Minopoli 1)	1982–84 source · spheroid (prolate spheroid (Yang), 2.9 km) 2018–23 shallow source · sill (penny-shaped crack (Fialko), 3.9 km) Deep sill-like reservoir (penny-shaped crack (Fialko), 8 km)	Arienzo 2026, Astort 2024, Cannatelli 2012, Gottsmann 2006, Pappalardo 2012, Petrelli 2023, Spera 2000
Kīlauea (Hawai‘i)	KIL_STORED — Degassed magma stored in HMM (2018) KIL_SUPPLY — Magma arriving at the summit reservoirs KIL_PRIMARY — CO_2-rich primary magma	HMM · prolate spheroid (prolate spheroid (Yang), 2.2 km) HMM · McTigue sphere (finite sphere (McTigue), 2 km) South Caldera · sill (penny-shaped crack (Fialko), 3.6 km) South Caldera · McTigue sphere (finite sphere (McTigue), 3 km)	Anderson 2019, Anderson 2020, Gansecki 2019, Gerlach 2002, Greenland 1985, Lerner 2021, Spera 2000, USGS 2022, Wang 2021

Etna (Italy)

Alkali basalt (trachybasalt) system of the 2002–03 flank eruption and the 2009–2021 recharge and lava-fountain cycles. Three magmas from the 2002 melt-inclusion record: the primitive volatile-rich basalt rising from ≥10 km b.s.l., the more evolved magma ponding at 4–6 km b.s.l., and the degassed shallow melt. Four sources: the GPS recharge and discharge centroids (McTigue spheres), the sill-like main chamber from fracture statistics, and the shallow lava-fountain spheroid.

Depths and pressures. The geodetic sources are published in km below sea level. For the flat half-space the free surface is placed 1.6 km above sea level (the free-surface elevation adopted for half-space event locations at Etna, Bonaccorso et al. 2021; an approximation of the mean elevation of the upper flanks), so depth_km = depth b.s.l. + 1.6 km. Pressures use P_depth_km = depth b.s.l. + 2.6 km with ρ = 2600 kg m⁻³, which reproduces Spilliaert et al. (2006)'s conversion of 170–220 MPa to 4–6 km b.s.l. (it accounts for the edifice above the sources).

Table 1 · Magmas (columns of the magma table, cell 5)

Quantity	Unit	ETNA_PRIMITIVE	ETNA_PONDING	ETNA_DEGASSED
description		Primitive 2002 basalt (lapilli, 21 Nov 2002)	Evolved 2002 magma ponding at 4–6 km b.s.l.	Degassed shallow melt (2002 matrix glass)
EVo solubility class		basalt	basalt	basalt
temperature	°C	1140	1125	1100
β_{liquid}	Pa^{-1}	1.25×10^{-10}	1.25×10^{-10}	1.25×10^{-10}
ρ_{melt}	kg m^{-3}	2650	2650	2650
H_2O (total)	wt%	3.4	2.5	1
CO_2 (total)	wt%	0.31	0.12	0.025
S (total)	wt%	0.32	0.31	0.13
redox	log units	NNO +0	NNO +0	NNO -0.5
Major elements, wt% (volatile-free; all Fe as FeO)				
SiO_2	wt%	47.15	48.16	50.04
TiO_2	wt%	1.62	1.62	2.06
Al_2O_3	wt%	15.65	18.23	16.22
FeO_{total}	wt%	10.46	9.63	10.48
MnO	wt%	0.18	0.17	0.23
MgO	wt%	7.53	4.65	3.23
CaO	wt%	11.73	9.92	7.43
Na_2O	wt%	2.97	4.03	4.41
K_2O	wt%	1.75	2.19	3.5
P_2O_5	wt%	0.44	0.55	1.04
total	wt%	99.48	99.15	98.64

Table 2 · Where each magma value comes from

Magma	Quantity	Source	Basis / note
ETNA_PRIMITIVE	major elements	Spilliaert 2006	Table 1, 21 Nov 2002 lapilli (south vents); Fe2O3(t) 11.62 → FeO(t) = 0.8998 × Fe2O3
	temperature	Spilliaert 2006	degassing model temperature 1140 °C (MELTS liquidus 1160 → 1125 °C)
	H_2O	Spilliaert 2006	primitive inclusions 3.4 ± 0.2 wt% (max 3.6)
	CO_2	Spilliaert 2006	maximum measured 3145 ppm; a minimum for the total budget
	S	Spilliaert 2006	primitive inclusions 0.27–0.35 wt% (max 0.347)
	redox (fO_2)	Spilliaert 2006	≈ NNO (after Métrich & Clocchiatti 1996); Gennaro et al. (2020): NNO−0.9 to +0.4 for 2002–2013 inclusions
	β_{liquid}	Scudero 2019	$\beta m \approx 1.25 \times 10^{-10}$ Pa^{-1} for Etnean basaltic magma
	ρ_{melt}	Spera 2000	typical basaltic melt density (used only if undersaturated)
ETNA_PONDING	major elements	Spilliaert 2006	Table 1, 4 Nov 2002 lava (north vents), the most evolved 2002 bulk rock
	temperature	Spilliaert 2006	lower end of the MELTS crystallisation interval (1125 °C)
	H_2O	Spilliaert 2006	ponding-cluster inclusions at 170–220 MPa: 2.2–2.8 wt%
	CO_2	Spilliaert 2006	ponding-cluster inclusions: 1000–1340 ppm
	S	Spilliaert 2006	ponding-cluster inclusions: 0.30–0.32 wt%
	redox (fO_2)	Spilliaert 2006	≈ NNO, as for the primitive magma
	β_{liquid}	Scudero 2019	as above
	ρ_{melt}	Spera 2000	as above
ETNA_DEGASSED	major elements	Spilliaert 2006	Table 2, matrix glass ET02-5 (29 Oct 2002), Fe as FeO
	temperature	assumption / derived	assumed, below the 1125 °C end of the MELTS interval
	H_2O	Spilliaert 2006	evolved glass embayments trapped at 25–50 MPa: 0.3–1.3 wt%
	CO_2	Spilliaert 2006	embayments: 140–270 ppm
	S	Spilliaert 2006	embayments: 0.11–0.14 wt%
	redox (fO_2)	Gennaro 2020	2008 and 2013 inclusions NNO−0.9 to −0.2
	β_{liquid}	Scudero 2019	as above
	ρ_{melt}	Spera 2000	as above

Table 3 · Reservoirs (rows of the reservoir table, cell 6)

Reservoir	Shape	Depth (km)	V_0 (m³)	V_e (m³)	Resident	Injected	μ (GPa)	ν	Radius (km)	b/a	Dip (°)	Azim. (°)	P datum (km)	ρ_c (kg/m³)
Recharge source · McTigue	finite sphere (McTigue)	8.1	4.19×10^{9}	5×10^{6}; 1.5×10^{7}; 3×10^{7}	ETNA_PRIMITIVE	ETNA_PRIMITIVE	20	0.25	1	—	—	—	9.1	2600
Discharge / ponding source · McTigue	finite sphere (McTigue)	6.1	4.19×10^{9}	1×10^{6}; 5×10^{6}; 1×10^{7}	ETNA_PONDING	ETNA_PRIMITIVE	20	0.25	1	—	—	—	7.1	2600
Main chamber · sill	penny-shaped crack (Fialko)	6.1	1.03×10^{11}	1×10^{7}; 3×10^{7}; 1×10^{8}	ETNA_PONDING	ETNA_PRIMITIVE	20	0.25	5.7	—	—	—	7.1	2600
Lava-fountain source · gas-rich	prolate spheroid (Yang)	2.35	3×10^{8}	1×10^{6}; 2×10^{6}; 5×10^{6}	ETNA_PONDING	ETNA_PRIMITIVE	20	0.25	—	0.5	90	0	3.35	2600
Lava-fountain source · degassed	prolate spheroid (Yang)	2.35	3×10^{8}	1×10^{6}; 2×10^{6}; 5×10^{6}	ETNA_DEGASSED	ETNA_PRIMITIVE	20	0.25	—	0.5	90	0	3.35	2600

Table 4 · Where each reservoir value comes from

Reservoir	Source	Basis
Recharge source · McTigue	Aloisi 2018	McTigue finite sphere; inflation centroid ≈ 6.5 km b.s.l. (phases 6.1–7.5 km); ΔV = 3×10^{6}–2×10^{7} m³
	assumption / derived	radius 1 km assumed (not reported); V0 = 4/3·π·(1 km)³
	Scudero 2019	E = 50 GPa, ν = 0.25 → μ = 20 GPa (Aloisi et al. 2018 used μ = 30 GPa, called an overestimate)
Discharge / ponding source · McTigue	Aloisi 2018	deflation centroid ≈ 4.5 km b.s.l.; ΔV = -1.6×10^{6} to -8.2×10^{6} m³
	Spilliaert 2006	ponding of the 2002 magma at 4–6 km b.s.l. (170–220 MPa)
Main chamber · sill	Scudero 2019	sill-like chamber 1 km thick, radius 5.7 km, most likely volume 103 km³ (range 69–206 km³), roof at 3–6 km b.s.l.; centre placed at 4.5 km b.s.l.
Lava-fountain source · gas-rich	Bonaccorso 2021	vertical prolate spheroid, aspect 0.5, 0.5–1 km b.s.l. (centre at 0.75 km), ΔV ≈ 2×10^{6} m³ per fountain, mean wall displacement ≈ 0.8 m
	Bonaccorso 2013	2011–2012 fountains: elongated source near sea level, ΔV ≈ 2×10^{6} m³
	assumption / derived	V0 = 3×10^{8} m³ derived so that ΔV/(wall area) ≈ 0.8 m for this shape
Lava-fountain source · degassed	assumption / derived	same geometry, filled with the degassed melt: shows how much the gas content of the resident magma matters

Table 5 · Check: saturation pressures computed by EVo from these inputs

Magma	EVo saturation pressure P_{sat} (MPa)	Independent petrological constraint
ETNA_PRIMITIVE	359.7	primitive inclusions trapped at ≈ 200–425 MPa (Spilliaert et al. 2006)
ETNA_PONDING	193.7	ponding cluster 170–220 MPa (Spilliaert et al. 2006)
ETNA_DEGASSED	44.0	evolved embayments trapped at 25–50 MPa (Spilliaert et al. 2006)

Table 6 · Result of the chain with these inputs (cell 11 of the notebook)

Reservoir	P (MPa)	Resident	Vapour present?	φ (vol%)	β_c (Pa^{-1})	r_V	Fraction of V_e seen at the surface
Recharge source · McTigue	232.1	ETNA_PRIMITIVE	yes	0.93	3.75×10^{-11}	5.37	19%
Discharge / ponding source · McTigue	181.1	ETNA_PONDING	yes	0.07	3.75×10^{-11}	4.44	23%
Main chamber · sill	181.1	ETNA_PONDING	yes	0.07	2.24×10^{-10}	1.58	63%
Lava-fountain source · gas-rich	85.4	ETNA_PONDING	yes	2.58	4.14×10^{-11}	11.24	9%
Lava-fountain source · degassed	85.4	ETNA_DEGASSED	no	0.00	4.14×10^{-11}	4.02	25%

Campi Flegrei (Italy)

Potassic caldera system: a trachytic (Agnano–Monte Spina-type) magma in the shallow storage zone and a trachybasaltic (shoshonitic) parental magma in the deep sill-like reservoir. Three sources: the 1982–84 unrest spheroid, the 2018–2023 shallow source and the caldera-wide deep sill.

Depths and pressures. Depths below the ground surface as published (Pozzuoli area, near sea level). Pressures ρgz with ρ = 2650 kg m⁻³ (the upper-crust density used by Arienzo et al. 2026 to convert pressures to depths).

Table 1 · Magmas (columns of the magma table, cell 5)

Quantity	Unit	CF_TRACHYTE	CF_TRACHYBASALT
description		Trachyte, Agnano–Monte Spina type (4.1 ka)	Trachybasalt (shoshonitic parental melt, Minopoli 1)
EVo solubility class		phonolite	basalt
temperature	°C	950	1160
β_{liquid}	Pa^{-1}	1.1×10^{-10}	8×10^{-11}
ρ_{melt}	kg m^{-3}	2400	2550
H_2O (total)	wt%	3	2.8
CO_2 (total)	wt%	0.05	0.11
S (total)	wt%	0.02	0.1
redox	log units	FMQ +1	FMQ +1
Major elements, wt% (volatile-free; all Fe as FeO)			
SiO_2	wt%	61.67	47.96
TiO_2	wt%	0.49	0.78
Al_2O_3	wt%	18.63	14.61
FeO_{total}	wt%	3.38	7.73
MnO	wt%	0.01	0.08
MgO	wt%	0.67	9.37
CaO	wt%	2.48	8.94
Na_2O	wt%	4.13	1.57
K_2O	wt%	8.54	3.92
P_2O_5	wt%	—	0.56
total	wt%	100.00	95.52

Table 2 · Where each magma value comes from

Magma	Quantity	Source	Basis / note
CF_TRACHYTE	major elements	Pappalardo 2012	Table 2, Agnano–Monte Spina unit A matrix glass (P2O5 below detection)
	temperature	Arienzo 2026	950 °C adopted for the trachytic melt inclusions
	H_2O	Petrelli 2023	AMS melt inclusions 0.8–3.0 wt% (upper end used); Arienzo et al. (2026): 0.5–2.9 wt% for all CF inclusions
	CO_2	Petrelli 2023	AMS inclusions 150–500 ppm (upper end used); Arienzo et al. (2026) report up to 2666 ppm in AMS-A inclusions
	S	Arienzo 2026	S 200–801 ppm, lowest in AMS
	redox (fO_2)	Cannatelli 2012	QFM to QFM+1 for Campi Flegrei magmas; assumed FMQ+1 (no direct measurement for the trachyte)
	β_{liquid}	Spera 2000	evolved alkaline melt (thesis value for trachyte)
	ρ_{melt}	Spera 2000	trachytic melt density (used only if undersaturated)
CF_TRACHYBASALT	major elements	Cannatelli 2012	parental melt inclusion Mi1-C1 o6 M1 (Minopoli 1)
	temperature	Cannatelli 2012	clinopyroxene saturation of the Minopoli 1 parental melt (liquidus 1300 °C)
	H_2O	Petrelli 2023	Minopoli 2 / Fondo Riccio inclusions 0.2–2.84 wt% (upper end); Cannatelli et al.: 1–4 wt%
	CO_2	Petrelli 2023	Minopoli 2 / Fondo Riccio inclusions 172–1100 ppm (upper end)
	S	Cannatelli 2012	parental melt SO2 0.21 wt% → S 0.105 wt%
	redox (fO_2)	Cannatelli 2012	QFM+1 for Minopoli 1
	β_{liquid}	Spera 2000	mafic melt (thesis value for shoshonite)
	ρ_{melt}	Spera 2000	mafic potassic melt (used only if undersaturated)

Table 3 · Reservoirs (rows of the reservoir table, cell 6)

Reservoir	Shape	Depth (km)	V_0 (m^3)	V_e (m^3)	Resident	Injected	μ (GPa)	ν	Radius (km)	b/a	Dip (°)	Azim. (°)	P datum (km)	ρ_c (kg/m^3)
1982–84 source · spheroid	prolate spheroid (Yang)	2.9	1.16×10^{10}	1×10^{7}; 7.1×10^{7}; 1×10^{8}	CF_TRACHYTE	CF_TRACHYBASALT	10	0.25	—	0.51	90	0	—	2650
2018–23 shallow source · sill	penny-shaped crack (Fialko)	3.9	1.6×10^{8}	1×10^{6}; 1×10^{7}; 6×10^{7}	CF_TRACHYTE	CF_TRACHYBASALT	10	0.25	2	—	—	—	—	2650
Deep sill-like reservoir	penny-shaped crack (Fialko)	8	1×10^{11}	1×10^{7}; 1×10^{8}; 1×10^{9}	CF_TRACHYBASALT	CF_TRACHYBASALT	10	0.25	5.64	—	—	—	—	2650

Table 4 · Where each reservoir value comes from

Reservoir	Source	Basis
1982–84 source · spheroid	Gottsmann 2006	vertical prolate spheroid 800 m E of Pozzuoli, depth 2.9 km (2.0–4.2), aspect 0.51, semi-major axis 2.2 km; $\Delta V = 71\times10^6$ m³ and $\Delta P = 52$ MPa for 1982–84; $\mu = 10$ GPa, $\nu = 0.25$
	assumption / derived	$V0 = 4/3\cdot\pi\cdot a\cdot b^2$ with a = 2.2 km, b/a = 0.51
2018–23 shallow source · sill	Astort 2024	shallow source at 3.9 km (2018–2023), sill-like to oblate; volume ≈ 0.16 km³ in 2023; volume change 1.4–8.3×10⁶ m³/yr, > 6×10⁷ m³ in 2007–2023
	assumption / derived	crack radius 2 km assumed (not reported): thin ($\bar{w}/2a \approx 0.003$) and buried ($a/d \approx 0.5$)
	Gottsmann 2006	$\mu = 10$ GPa, $\nu = 0.25$
Deep sill-like reservoir	Astort 2024	caldera-wide tabular source fixed at 8 km, area ≈ 100 km² (radius 5.64 km)
	Pappalardo 2012	sill-like magma layer 1–2 km thick at 7–9 km; 1 km thickness used → V0 = 100 km³
	Petrelli 2023	main reservoir ≈ 9 km

Table 5 · Check: saturation pressures computed by EVo from these inputs

Magma	EVo saturation pressure P_{sat} (MPa)	Independent petrological constraint
CF_TRACHYTE	330.0	AMS inclusions 107–211 MPa (Arienzo et al. 2010; Petrelli et al. 2023): EVo's P_{sat} is higher, so this budget is an upper bound
CF_TRACHYBASALT	191.3	Minopoli 2 / Fondo Riccio inclusions 61–212 MPa (Petrelli et al. 2023)

Table 6 · Result of the chain with these inputs (cell 11 of the notebook)

Reservoir	P (MPa)	Resident	Vapour present?	φ (vol%)	β_c (Pa^{-1})	r_V	Fraction of V_e seen at the surface
1982–84 source · spheroid	75.4	CF_TRACHYTE	yes	1.39	8.25×10^{-11}	4.55	22%
2018–23 shallow source · sill	101.4	CF_TRACHYTE	yes	0.55	1.06×10^{-8}	1.02	98%
Deep sill-like reservoir	208.0	CF_TRACHYBASALT	no	0.00	4.04×10^{-10}	1.20	83%

Kīlauea (Hawai‘i)

Tholeiitic basalt summit system with two reservoirs: Halema‘uma‘u (HMM, ~2 km) and South Caldera (SC, ~3–4 km). Three volatile states of the same basalt: the CO_2-rich primary magma, the magma arriving at the summit reservoirs, and the degassed magma stored in HMM before 2018. Each reservoir is given in the two published geometries.

Depths and pressures. Depths below the caldera floor as published. Pressures ρgz with ρ = 2500 kg m^{-3}, which gives ≈ 49 MPa at 2 km, close to the ≈ 45 MPa magmastatic pressure of the HMM reservoir (Anderson et al. 2019).

Table 1 · Magmas (columns of the magma table, cell 5)

Quantity	Unit	KIL_STORED	KIL_SUPPLY	KIL_PRIMARY
description		Degassed magma stored in HMM (2018)	Magma arriving at the summit reservoirs	CO_2-rich primary magma
EVo solubility class		basalt	basalt	basalt
temperature	°C	1150	1200	1220
β_{liquid}	Pa^{-1}	8×10^{-11}	8×10^{-11}	8×10^{-11}
ρ_{melt}	kg m^{-3}	2600	2600	2600
H_2O (total)	wt%	0.27	0.32	0.32
CO_2 (total)	wt%	0.0265	0.32	0.7
S (total)	wt%	0.111	0.09	0.09
redox	log units	FMQ +0.25	FMQ +0.25	FMQ +0.25
Major elements, wt% (volatile-free; all Fe as FeO)				
SiO_2	wt%	49.3	49.3	49.3
TiO_2	wt%	2.78	2.78	2.78
Al_2O_3	wt%	13.6	13.6	13.6
FeO_{total}	wt%	11.3	11.3	11.3
MnO	wt%	0.17	0.17	0.17
MgO	wt%	7.13	7.13	7.13
CaO	wt%	11.4	11.4	11.4
Na_2O	wt%	2.4	2.4	2.4
K_2O	wt%	0.51	0.51	0.51
P_2O_5	wt%	0.29	0.29	0.29
total	wt%	98.88	98.88	98.88

Table 2 · Where each magma value comes from

Magma	Quantity	Source	Basis / note
KIL_STORED	major elements	USGS 2022	BHVO-2(G), 1919 Halema‘uma‘u overflow lava; K2O from K = 4270 ppm
	temperature	Gansecki 2019	Halema‘uma‘u lava lake 1155–1160 °C; Lerner et al. (2021): 1130–1145 °C on eruption
	H_2O	Lerner 2021	best pre-eruptive estimate 0.27 ± 0.03 wt%
	CO_2	Lerner 2021	≈ 265 ppm (minimum pre-eruptive, interstitial melt 224 ppm)
	S	Lerner 2021	pre-eruptive 1110 ± 60 ppm
	redox (fO_2)	Lerner 2021	$Fe^{3+}/\Sigma Fe$ = 0.18; FMQ+0.25
	β_{liquid}	Spera 2000	basaltic melt
	ρ_{melt}	Anderson 2019	reservoir magma 2550–2650 kg m^{-3}
KIL_SUPPLY	major elements	USGS 2022	BHVO-2(G), 1919 Halema‘uma‘u overflow lava; K2O from K = 4270 ppm
	temperature	assumption / derived	assumed, between eruption (1140–1160 °C) and high-Fo inclusion (1220–1310 °C) temperatures
	H_2O	Greenland 1985	magma arriving at the shallow summit reservoir: 0.32 wt% H2O
	CO_2	Greenland 1985	0.32 wt% CO2
	S	Greenland 1985	0.09 wt% S
	redox (fO_2)	Lerner 2021	as above
	β_{liquid}	Spera 2000	basaltic melt
	ρ_{melt}	Anderson 2019	as above
KIL_PRIMARY	major elements	USGS 2022	BHVO-2(G), 1919 Halema‘uma‘u overflow lava; K2O from K = 4270 ppm (no primary-melt analysis used)
	temperature	Lerner 2021	high-Fo melt-inclusion entrapment 1220–1310 °C (lower end)
	H_2O	Greenland 1985	as for the supply magma
	CO_2	Gerlach 2002	bulk CO2 of the primary magma ≈ 0.70 wt%
	S	Greenland 1985	as for the supply magma
	redox (fO_2)	Lerner 2021	as above
	β_{liquid}	Spera 2000	basaltic melt
	ρ_{melt}	Anderson 2019	as above

Table 3 · Reservoirs (rows of the reservoir table, cell 6)

Reservoir	Shape	Depth (km)	V_0 (m^3)	V_e (m^3)	Resident	Injected	μ (GPa)	ν	Radius (km)	b/a	Dip (°)	Azim. (°)	P datum (km)	ρ_c (kg/m^3)
HMM · prolate spheroid	prolate spheroid (Yang)	2.2	4×10^9	3.5×10^5; 1×10^7; 1×10^8	KIL_STORED	KIL_SUPPLY	3	0.25	—	0.56	90	0	—	2500
HMM · McTigue sphere	finite sphere (McTigue)	2	4×10^9	3.5×10^5; 1×10^7; 1×10^8	KIL_STORED	KIL_SUPPLY	3	0.25	—	—	—	—	—	2500
South Caldera · sill	penny-shaped crack (Fialko)	3.6	2×10^9	3.5×10^5; 1×10^7; 1×10^8	KIL_SUPPLY	KIL_PRIMARY	3	0.25	1.5	—	—	—	—	2500
South Caldera · McTigue sphere	finite sphere (McTigue)	3	2.14×10^9	3.5×10^5; 1×10^7; 1×10^8	KIL_SUPPLY	KIL_PRIMARY	3	0.25	0.8	—	—	—	—	2500

Table 4 · Where each reservoir value comes from

Reservoir	Source	Basis
HMM · prolate spheroid	Wang 2021	HMM prolate spheroid, vertical/horizontal aspect ≈ 1.8 (b/a = 0.56), depth ≈ 2.2 km
	Anderson 2019	reservoir volume 2.5–7.2 km^3 (68%); 4 km^3 used; μ prior 2–10 GPa, 3 GPa in earlier work
	Anderson 2020	deflation–inflation events ≈ 3.5×10^5 m^3
HMM · McTigue sphere	Anderson 2019	reservoir centred ≈ 2 km below the surface, roughly equant; 4 km^3 (radius 0.98 km from V0)
South Caldera · sill	Wang 2021	SC oblate reservoir, aspect ≈ 0.14, depth ≈ 3.6 km
	assumption / derived	radius 1.5 km assumed: an oblate body of this radius and c/a = 0.14 has V ≈ 2.0 km^3, the volume of the 0.8-km sphere of Anderson et al. (2020)
South Caldera · McTigue sphere	Anderson 2020	SC reservoir as a McTigue sphere at 3 km, radius 800 m

Table 5 · Check: saturation pressures computed by EVo from these inputs

Magma	EVo saturation pressure P_{sat} (MPa)	Independent petrological constraint
KIL_STORED	35.5	2018 inclusions mostly 2–100 MPa, bubble-free ones ≈ 58 MPa (Lerner et al. 2021)
KIL_SUPPLY	314.4	no inclusion constraint; Gerlach et al. (2002) find most CO_2 already exsolved at summit-reservoir depths, consistent with P_{sat} far above the 50–90 MPa of the reservoirs
KIL_PRIMARY	505.0	no inclusion constraint; the CO_2-rich primary magma is expected to saturate well below the summit reservoirs (Gerlach et al. 2002)

Table 6 · Result of the chain with these inputs (cell 11 of the notebook)

Reservoir	P (MPa)	Resident	Vapour present?	φ (vol%)	β_c (Pa^{-1})	r_V	Fraction of V_e seen at the surface
HMM · prolate spheroid	54.0	KIL_STORED	no	0.00	2.71×10^{-10}	1.30	77%
HMM · McTigue sphere	49.0	KIL_STORED	no	0.00	2.5×10^{-10}	1.32	76%
South Caldera · sill	88.3	KIL_SUPPLY	yes	2.17	1.16×10^{-9}	1.28	78%
South Caldera · McTigue sphere	73.6	KIL_SUPPLY	yes	2.72	2.5×10^{-10}	2.79	36%

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Limitations

• These presets are physically consistent starting points, not unique reconstructions: each volcano has several published, partly conflicting estimates, and the choice made here is stated in Tables 2 and 4.

• Where a geometric parameter was not reported (Etna recharge radius, Campi Flegrei shallow-sill radius, Kīlauea South Caldera radius) the value is an explicit assumption; r_V of spheres and spheroids does not depend on the reservoir volume, that of cracks does.

• Melt-inclusion volatiles are lower bounds of pre-eruptive budgets; raising CO_2 raises the saturation pressure and the vapour fraction at storage depth.

• The half-space ignores topography and elastic layering; at Etna the edifice is taken into account only through the pressure datum.